



Honeybee Swarm Dynamics: Investigating the Relationship
Between Individual Decision Making and Collective Foraging

Intro

Honeybees are social insects that use pheromone signals to direct each other, resulting in an emergent swarm behavior.

Little work has been done to study how the behavior of the bees changes when in the presence of the swarm.



https://en.wikipedia.org/wiki/Western_honey_bee

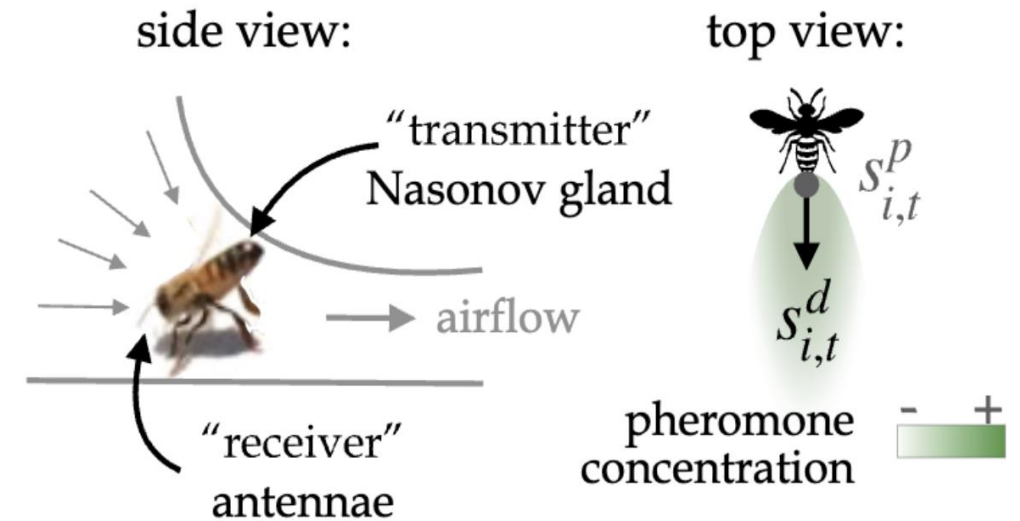
Our Goal

This study aims to improve the tracking of individual honeybees and investigate how the decisions of individual honeybees affect the foraging and aggregation process of the collective.

Literature Review

Previous work has shown that bees collectively create a scenting-mediated communication network by directional signaling away from the queen

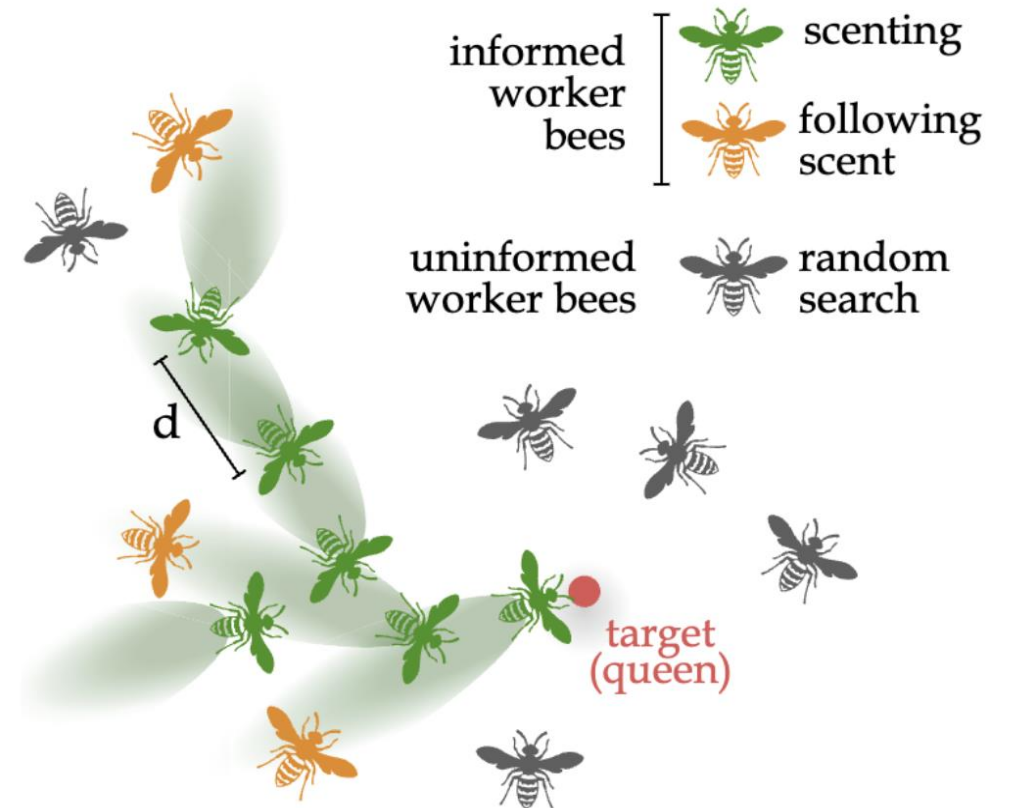
A Bees as chemical signal-amplifiers



Literature Review

Individual bees act as receivers and senders of signals by “scenting”, releasing pheromones from the Nasonov glands and fanning their wings to direct the signals backward

B Solution to the localization problem



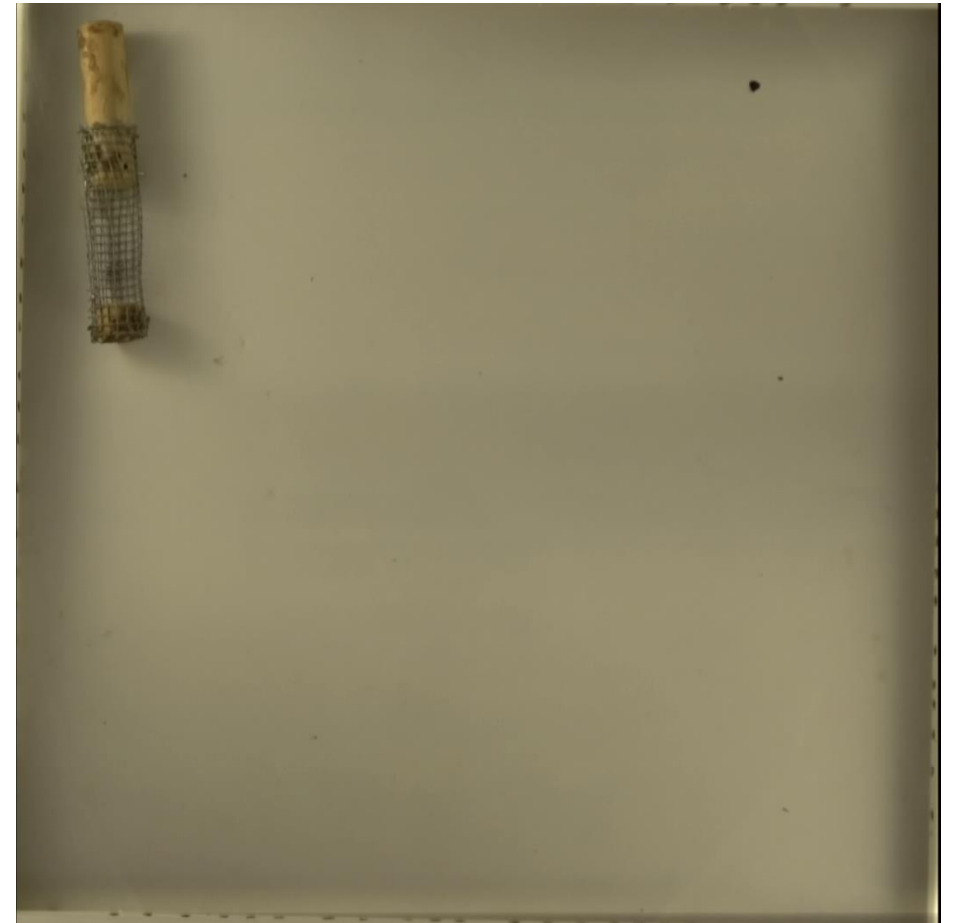
Hypothesis

The time-series data from individual honeybees will display a predictable relationship between scenting and exploring, allowing for collective foraging

Research Methods

Established a behavioral assay in which worker bees search for a stationary caged queen in a semi-two-dimensional arena

Use an object tracking and segmentation algorithm to gather time-series data on the scenting behaviors of individual bees when separated from their queen



Our First Attempt

XMem - Long-Term Video Object
Segmentation with an Atkinson-
Shiffrin Memory Model



Looks Promising

XMem is used in tools like
Segment Anything by Meta AI



The Issue with XMem

The system works well for complicated scenes with a lot of variety

Not so well for extremely little variety,

- The paper stated they were not able to distinguish between similar birds close together



The Fundamental Issue

We couldn't distinguish between
similarly shaped and colored objects
(when occluded)

Our Algorithm

Algorithm 1 Honeybee Tracking and Segmentation

```
background = get_background(video)
for frame ∈ video do
    processed = preprocess(frame, background)
    contours = form_contours(processed)
    tracks = create_tracks(contours, prev_tracks)
    groups = [group in tracks]
    if len(groups) > 0 then
        split_tracks = split_groups(tracks)
        tracks.remove(groups)
        tracks.extend(split_tracks)
    end if
    frame = draw_tracks(frame, tracks)
    show_frame(frame)
    prev_tracks = split_tracks
end for
```

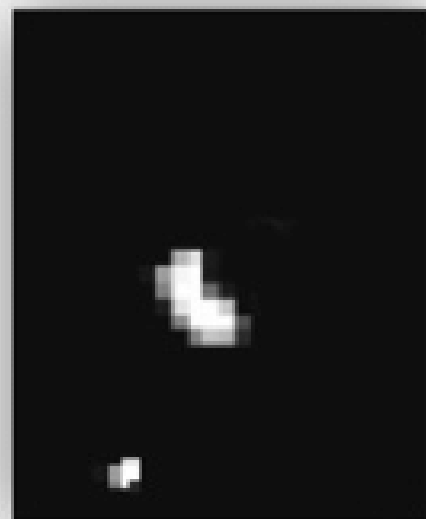
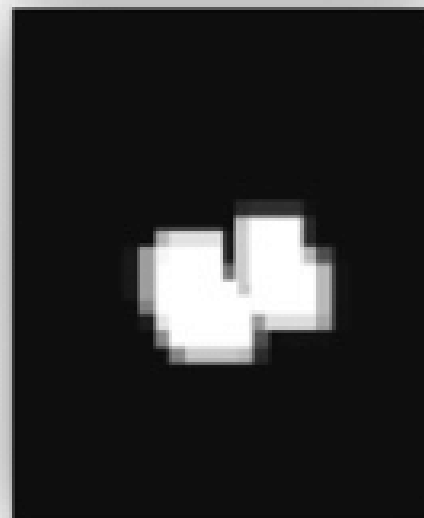
preprocess()

Median Background Subtraction

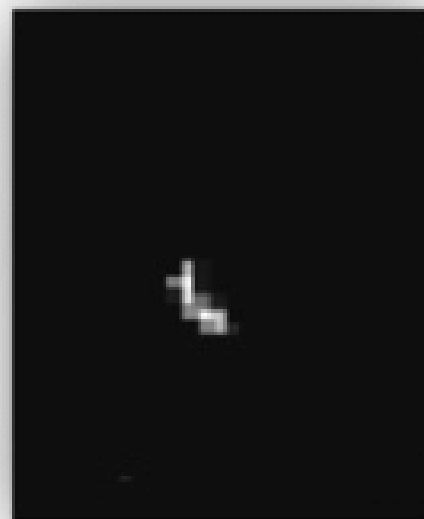




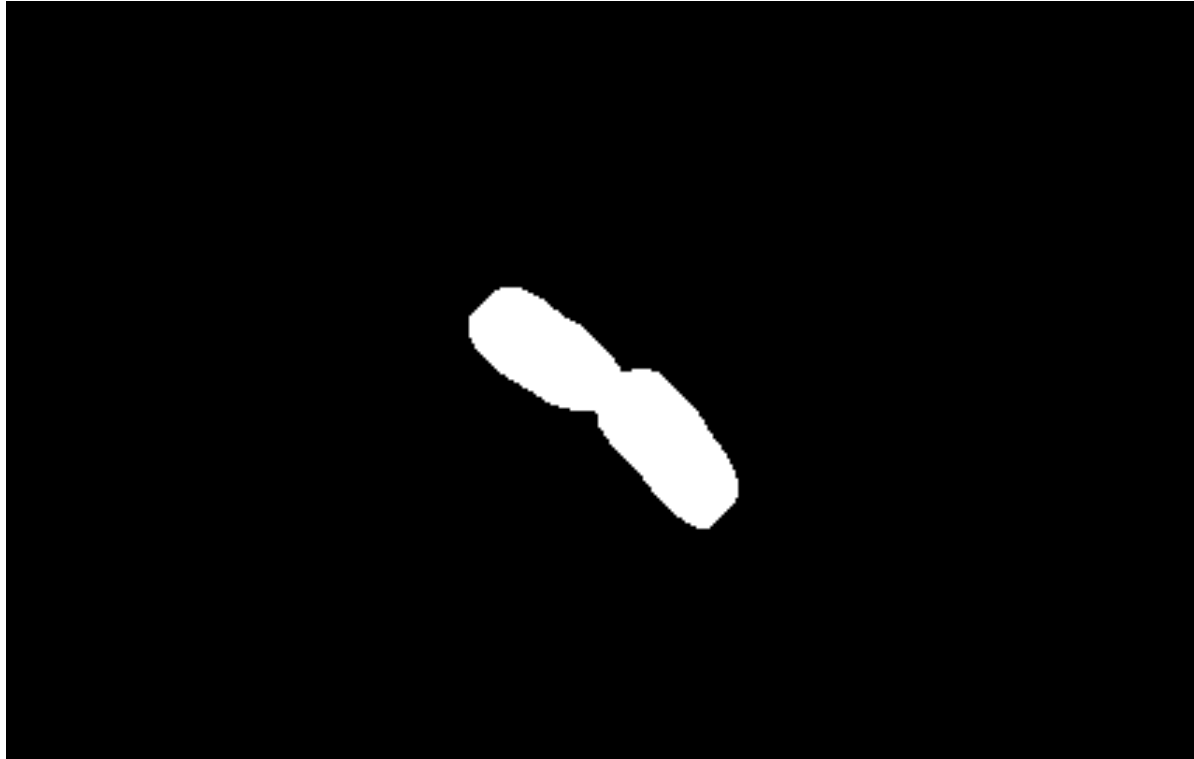
Dilation



Erosion

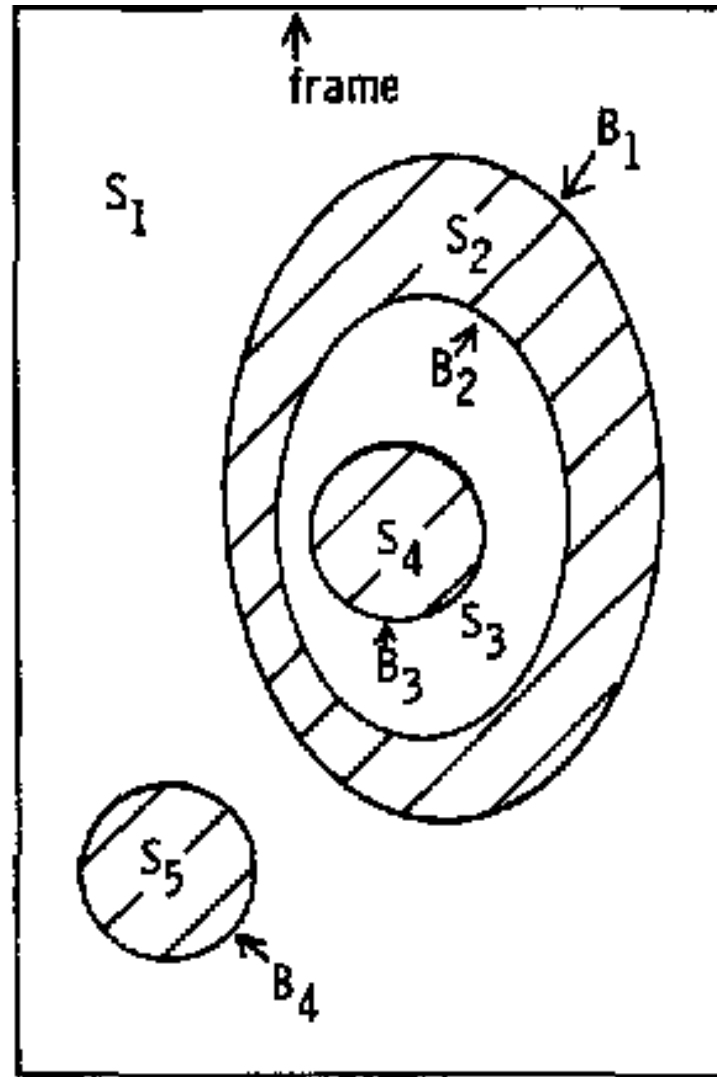


preprocess()

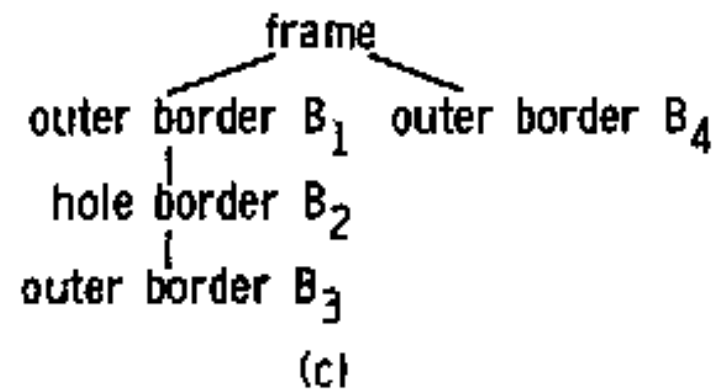
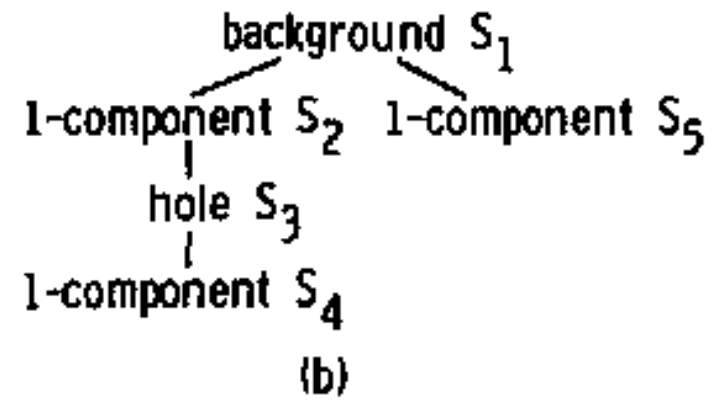


form_contours()





(a)



Suzuki, S., & Abe, K. (1985). Topological structural analysis of digitized binary images by border following. *Comput. Vis. Graph. Image Process.*, 30, 32-46.

`create_tracks()`



So, we're done right??

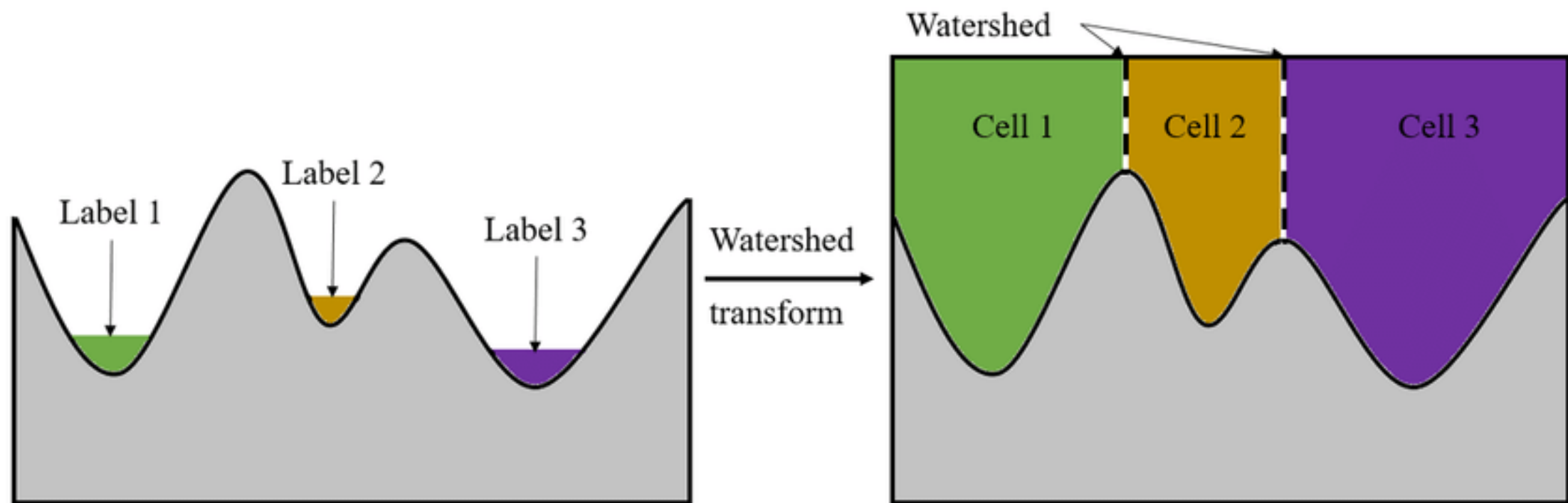
Not quite, there's an edge case (or two)



How to Separate Groups?

The watershed algorithm





split_groups()

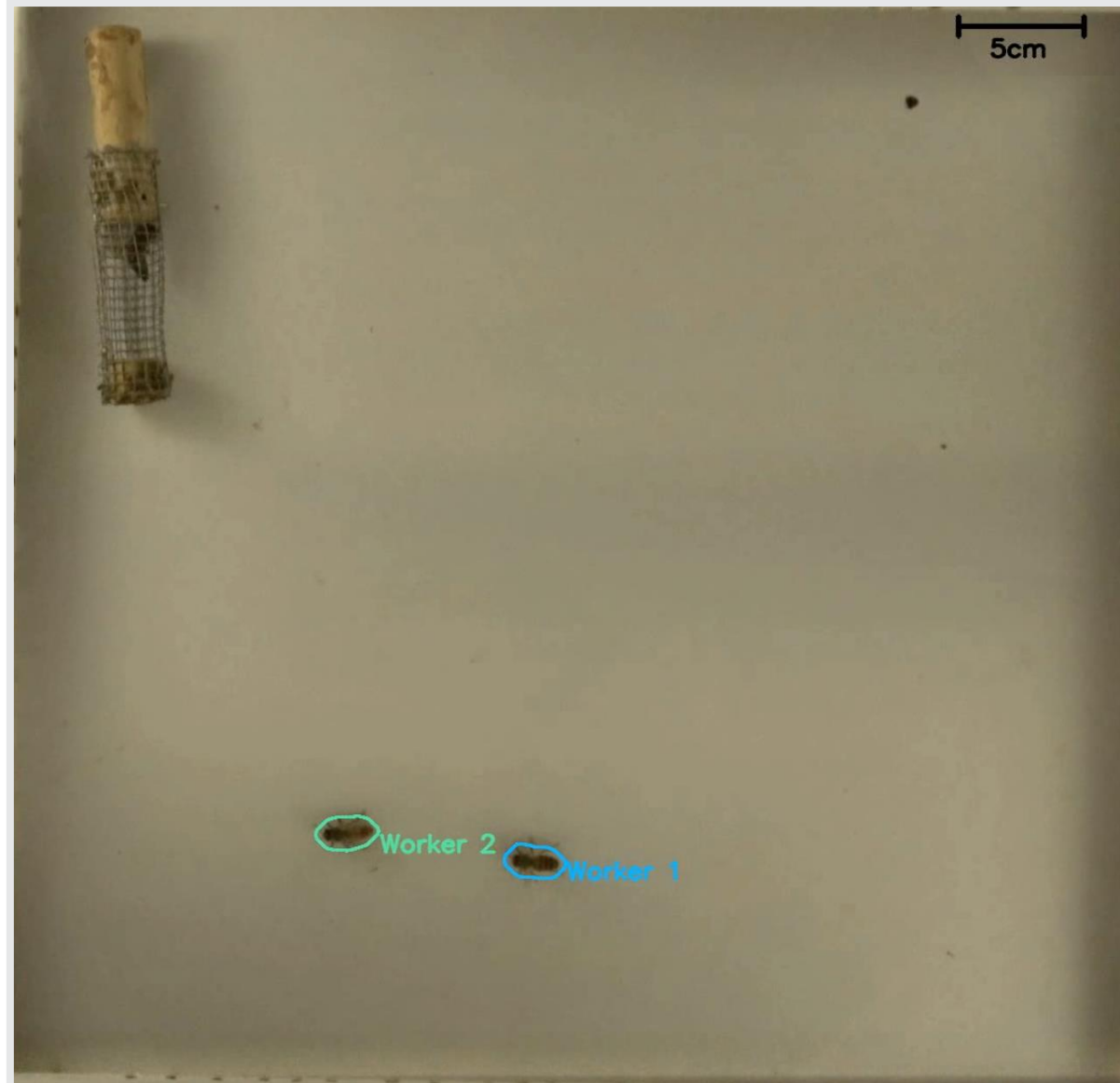


Additional Edge Cases

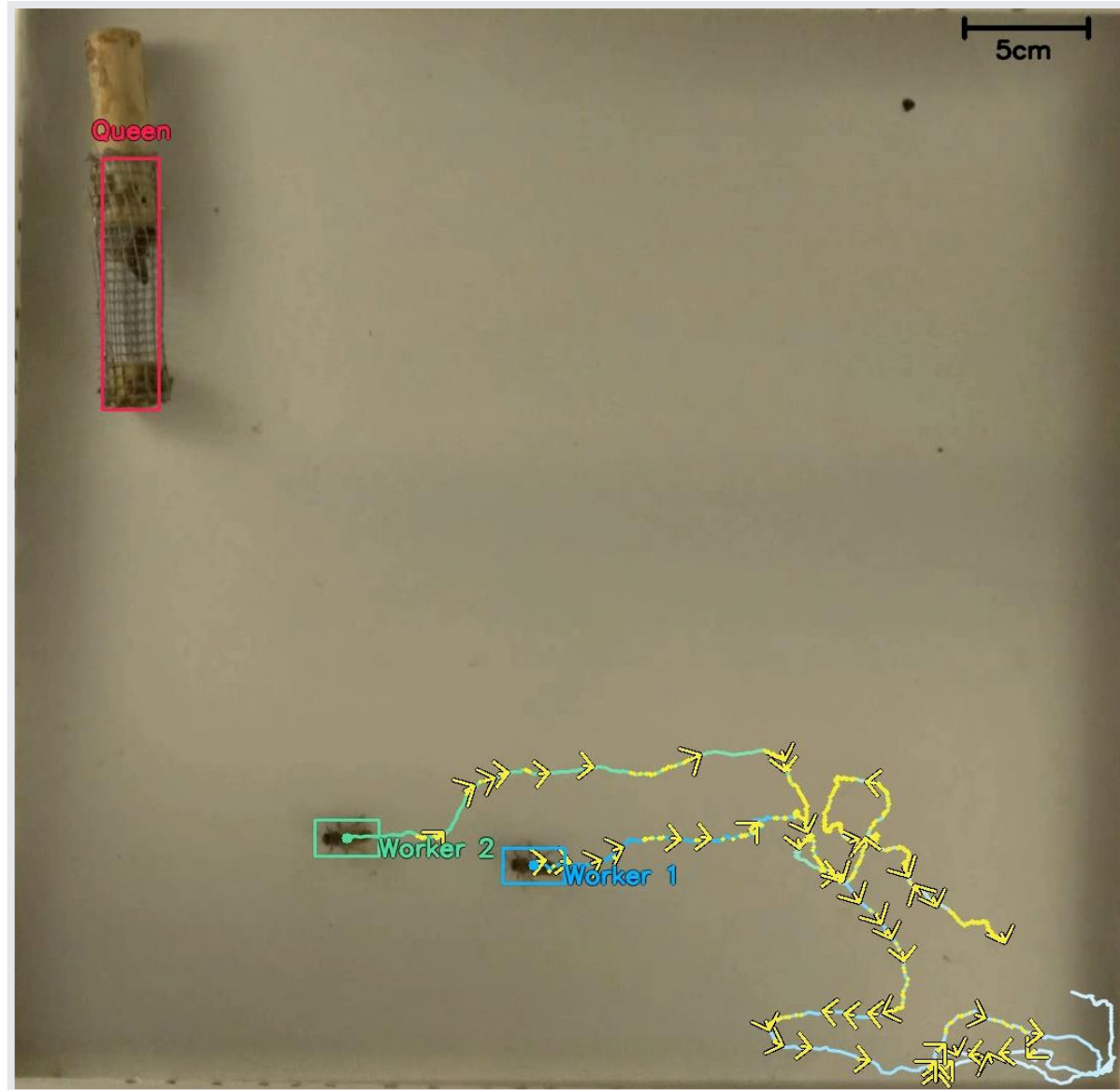
Color Thresholding
Manual Mode



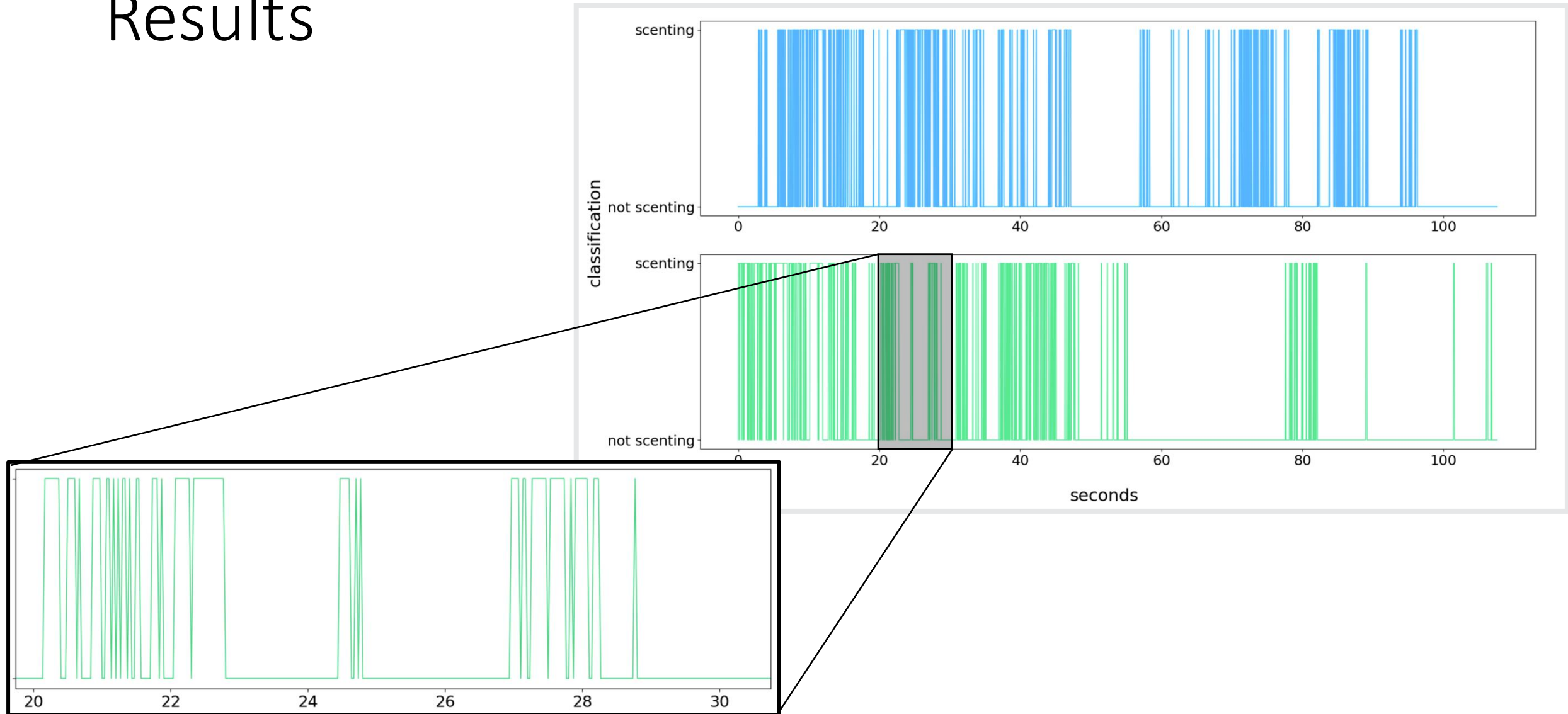
Results



Results



Results



5cm

Queen



Worker 2



Worker 1

Current Issues

Specific orientations with larger numbers of bees can cause the system to break down

- A bee from a previous frame could be matched with another close bee instead of the group

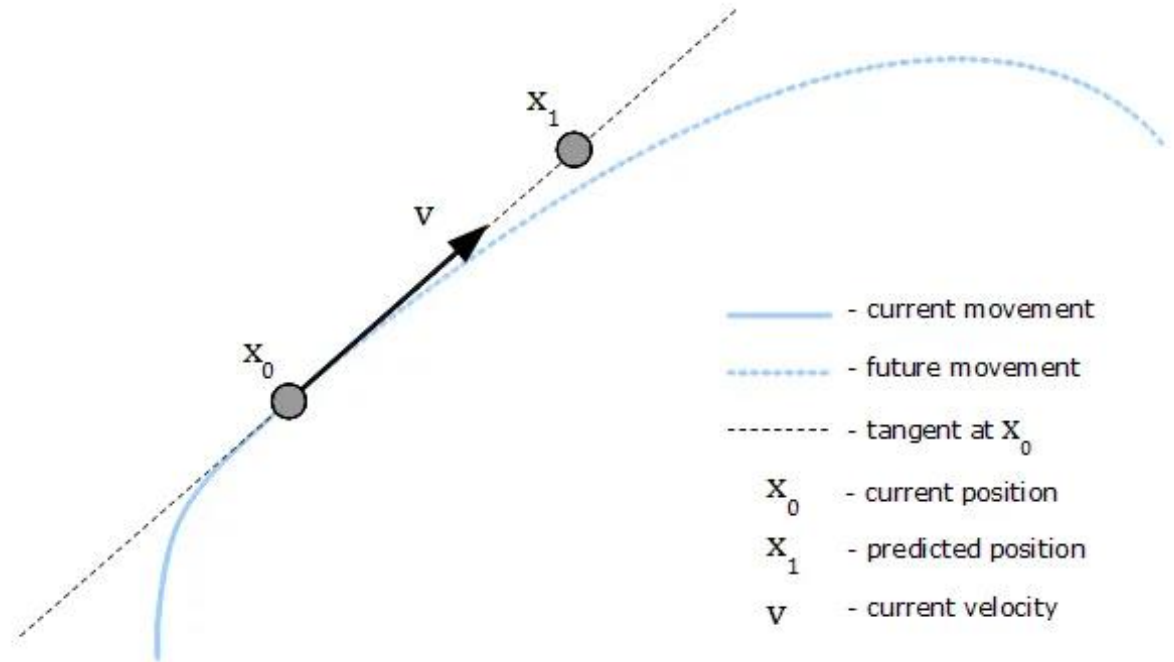


Potential Solution

Movement Prediction!

- Get prev direction of bee by getting the longest axis of its contour
- Take an extra step in direction based on prev velocity

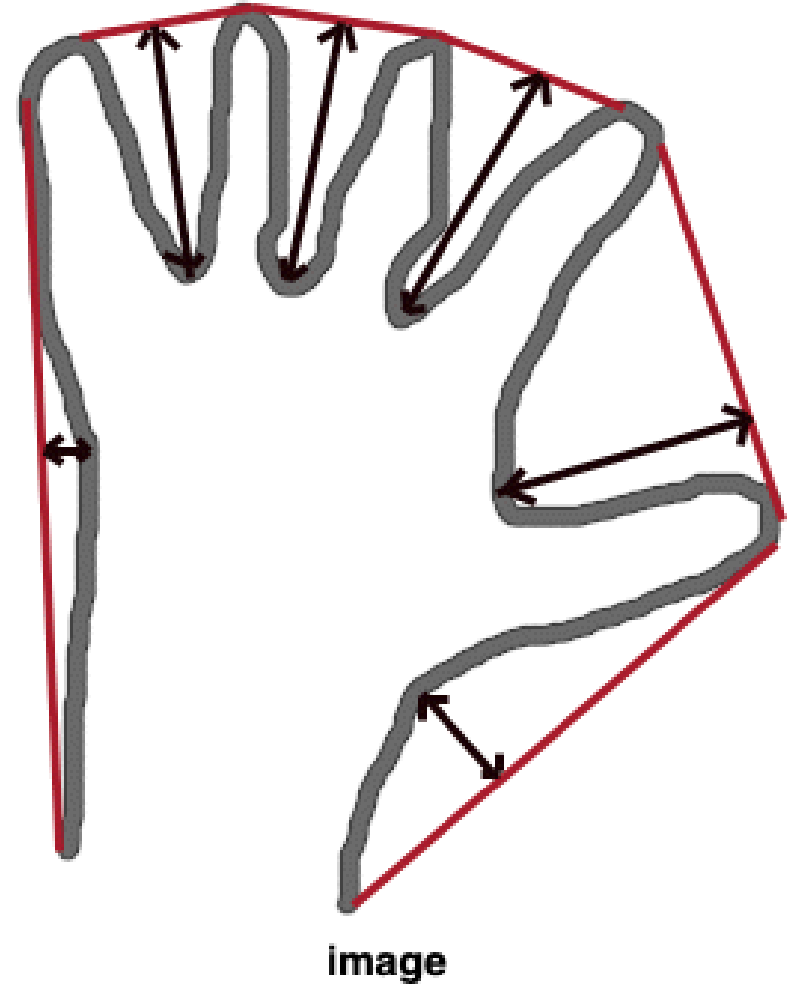
Cons: Bees don't always move consistently



Potential Solution

Convexity Defects

- Prev solution is matching prev center to closest current center
- Now match prev closest defect (or center if not grouped) to current closest defect
- Eliminates issue of the group's center being offset



Next Steps

Run similar tests with a larger number of bees and improve the tracking algorithm to require less human interaction

Visualize pheromone concentration gradients

Analyze resulting distributions and determine the correlation between the data for each trial

Acknowledgments

Javon Hickmon 1, Golnar Gharooni Fard 2, Orit Peleg 3,4

1 Department of Computer Science, University of Washington | 2 Department of Computer Science, University of Colorado Boulder 3 Bio Frontiers Institute, University of Colorado Boulder | 4 Santa Fe Institute

References

- [1] Nguyen, D. M. T., Iuzzolino, M. L., Mankel, A., Bozek, K., Stephens, G. J., & Peleg, O. (2021). Flow-mediated olfactory communication in honeybee swarms. *Proceedings of the National Academy of Sciences*, 118(13), e2011916118.
- [2] Nguyen, D. M. T., Fard, G. G., Atkins, A., Bontempo, P., Iuzzolino, M. L., & Peleg, O. (2023). Honey bees find the shortest path: a collective flow-mediated approach. *Artificial Life and Robotics*, 28(1), 1-7.

